Module 1 - Number Systems

Chapter 1- Numbers

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1. Find the place value of 4 in 46127.
   a) 4 b) 400 c) 40000 d) 4000 e) None of the above
2. Find the place value of 7 in 837218.
   a) 7000 b) 7 c) 700 d) 70000 e) None of the above
3. What is the place value of 6 in 65489203?
   a) 6×10^5 b) 6×10^4 c) 6×10^7 d) 6×10^8 e) None of the above
4. Find the face value of 7 in 942756.
   a) 7 b) 700 c) 7000 d) 70000
5. Find the face value of 6 in 652410.
   a) 6×10^5 b) 6×10^4 c) 6000 d) 6
6. Find the sum of the face values of 9 and 6 in 907364.
   a) 15 b) 20 c) 9 d) 18
7. Find the difference of the face values of 7 and 2 in 210978.
   a) 4 b) 3 c) 6 d) 5 e) None of the above
8. Find the sum of place and face values of 8 in 43836.
   a) 88 b) 808 c) 880 d) 888 e) None of the above
9. Find the difference of place and face values of 4 in 324372.
   a) 3996 b) 3998 c) 3398 d) 3396 e) None of the above
10. Find the sum of place value of 6 and face value of 9 in 927653.
    a) 608 b) 508 c) 609 d) 507 e) None of the above
11. Find the difference of place value of 4 and face value of 3 in 3784105.
    a) 3997 b) 1 c) 1000 d) 3845 e) None of the above
12. When 121012 is divided by 12, the remainder is
    a) 0 b) 2 c) 3 d) 4

13. The sum of place values of 2 in 2424 is

    a) 4 b) 220 c) 2002 d) 2020

14. The pair of numbers which are relatively prime to each other is

    a) (68, 85) b) (65, 91) c) (92, 85) d) (102, 153)

15. *Find the sum of 1st and 2nd prime numbers.

    a) 5 b) 3 c) 7 d) 2 e) None of the above

16. *Find the product of 1st natural number and 1st prime number.

    a) 4 b) 3 c) 2 d) 5 e) None of the above

17. *The product of 1st natural, 1st whole and the 1st prime numbers is equal to

    a) 5 b) 0 c) 9 d) 7 e) None of the above

18. *The product of any number and the 1st whole number is equal to

    a) 0 b) 2 c) 1 d) -1 e) None of the above

19. *A rational number is expressed as _____ where, p and q are integers and $q \neq 0$.

    a) pq b) $p + q$ c) $p - q$ d) p/q e) None of the above

20. *$2/3$ is a rational number whereas $\sqrt{2}/\sqrt{3}$ is

    a) also a rational number b) an irrational number c) not a number d) a natural periodic number

21. Which of the following is a prime number?

    a) 35 b) 53 c) 88 d) 90 e) None of the above

22. The number of all prime numbers less than 40 is _____

    a) 15 b) 18 c) 17 d) 12 e) None of the above

23. Find the quotient when 445 is divided by 5.

    a) 78 b) 48 c) 79 d) 89 e) None of the above

24. Find the remainder when 54 is divided by 17.

    a) 5 b) 3 c) 2 d) 7 e) None of the above

25. Find the dividend when divisor is 13, quotient is 30 and remainder is 12.

    a) 402 b) 543 c) 436 d) 455 e) None of the above

26. What is the remainder in the expression $29 \div 26$

    a) 3 b) 26 c) 29 d) 0 e) None of the above

27. Find the dividend from the expression $41\frac{4}{19}$

    a) 783 b) 800 c) 893 d) 387 e) None of the above

28. When 1/7 of a number is subtracted from the number itself, it gives the same value as the sum of all the angles of a triangle. What is the number?

    a) 224 b) 210 c) 140 d) 250 e) 187

29. What least number must be added to 1057 to get a number exactly divisible by 23?

    a) 1 b) 3 c) 2 d) 4 e) None of the above

30. In a division sum, the divisor is ten times the quotient and five times the remainder. If the remainder is 46, then find the dividend.

    a) 5388 b) 5343 c) 5336 d) 5391 e) None of the above

31. The product of two consecutive odd numbers is 6723. What is the greater number?

    a) 89 b) 85 c) 91 d) 83 e) None of the above

32. The sum of the four consecutive even numbers is 284. What would be the smallest number?

    a) 72 b) 74 c) 68 d) 66 e) None of the above

33. The sum of the digits of a two-digit number is 14 and the difference between the two digits of the number is 2. What is the product of the two digits of the two-digit number?

    a) 56 b) 48 c) 45 d) Couldn't be determined e) None of the above

34. What number should be added to 231228 to make it exactly divisible by 33?

    a) 1 b) 2 c) 3 d) 4

Higher Skill Level Questions

1. Find the sum of first 25 natural numbers.

    a) 432 b) 315 c) 325 d) 335 e) None of the above

2. Find the sum of the squares of first 35 natural numbers.

    a) 14910 b) 15510 c) 14510 d) 16510 e) None of the above

3. Find the sum of the cubes of first 15 natural numbers.

    a) 15400 b) 14400 c) 16800 d) 13300

4. Find the sum of first 37 odd numbers.

    a) 1369 b) 1295 c) 1388 d) 1875 e) None of the above

5. Find the sum of first 84 even numbers.

    a) 7140 b) 7540 c) 6720 d) 8832 e) None of the above

6. Sum of first 15 multiples of 8 is

   a) 960 b) 660 c) 1200 d) 1060

7. The product of four consecutive natural numbers plus one

   a) a non-square
   b) always sum of two square numbers
   c) a square
   d) None of the above

8. Find the unit digit in the product of $(268 \times 539 \times 826 \times 102)$.

   a) 5 b) 3 c) 4 d) 2 e) None of the above

9. Find the unit digit in the product of (4326×5321).

   a) 6 b) 8 c) 1 d) 3 e) None of the above

10. What is the unit digit in $(6817)^{754}$?

    a) 8 b) 4 c) 2 d) 9 e) None of the above

11. What is the unit digit in $(3^{65} \times 6^{59} \times 7^{71})$?

    a) 6 b) 4 c) 2 d) 1 e) None of the above

12. Find the last two-digits of $15 \times 37 \times 63 \times 51 \times 97 \times 17$

    a) 35 b) 45 c) 55 d) 85

13. How many rational numbers are there between 1 and 1000?

    a) 998 b) 999 c) 1000 d) Infinite

14. The sum of 5 consecutive even numbers A,B,C,D and E is 130. What is the product of A and E ?

    a) 720 b) 616 c) 660 d) 672 e) None of the above

15. The sum of the five consecutive numbers is equal to 170. What is the product of largest and the smallest numbers?

    a) 1512 b) 1102 c) 1152 d) 1210 e) None of the above

16. Which of the following numbers always divides the difference between the squares of two consecutive odd integers?

    a) 7 b) 3 c) 8 d) 6 e) None of the above

17. A number divided by 56 gives 29 as remainder. If the same number is divided by 8, the remainder will be _____

    a) 4 b) 5 c) 6 d) 7

18. On dividing a certain number by 357, the remainder is 39. On dividing the same number by 17, what will be the remainder?

    a) 5 b) 3 c) 7 d) 6 e) None of the above

19. A number when divided by 5, leaves 3 as remainder. What will be the remainder when the square of this number is divided by 5?

    a) 3 b) 4 c) 5 d) 0 e) None of the above

20. In a question on division with zero remainder, a candidate took 12 as divisor instead of 21. The quotient obtained by him was 35. Find the correct quotient.

    a) 10 b) 12 c) 20 d) 15 e) None of the above

21. A number when divided by a divisor leaves a remainder of 24. When twice the original number is divided by the same divisor, the remainder is 11. What is the value of the divisor?

    a) 13 b) 59 c) 35 d) 37

22. The number 58129745812974 is divisible by

    a) 11 b) 9 c) 4 d) None of the above

23. How many numbers between -11 and 11 are multiples of 2 or 3?

    a) 11 b) 14 c) 15 d) None of the above

24. Which one of the following numbers is divisible by 11?

    a) 45678940 b) 54857266 c) 87524398 d) 93455120 e) None of the above

25. When $17 * 200$ is divided by 18, find the remainder.

    a) 1 b) 4 c) 5 d) 3 e) None of the above

26. What is the remainder when 4^{1000} is divisible by 7?

    a) 1 b) 2 c) 4 d) None of the these

27. A common factor of $(41^{43} + 43^{43})$ and $(41^{41} + 43^{41})$ is _____

    a) $(43 - 41)$ b) $(41^{41} + 43^{41})$ c) $(41^{43} + 43^{43})$
    d) $(41 + 43)$ e) None of the above

28. The remainder when $9^{19} + 6$ is divided by 8 is

    a) 2 b) 3 c) 5 d) 7

29. What will be the remainder when 19^{100} is divided by 20?

    a) 19 b) 20 c) 3 d) 1

30. It is given that $(2^{32} + 1)$ is exactly divisible by a certain number. Which of the following is also definitely divisible by the same number?

a) $(2^{16} + 1)$ b) $(2^{16} - 1)$ c) 7×2^{13} d) $2^{96} + 1$ e) None of the above

31. The number $(6x^2 + 6x)$ for natural number x is always divisible by _____

a) 6 and 12 b) 12 only c) 6 only d) 3 only

32. $19^5 + 21^5$ is divisible by

a) Only 10 b) Only 20 c) Both 10 and 20 d) Neither 10 nor 20

33. If a is a natural number, then the largest number dividing $(a^3 - a)$ is

a) a b) 5 c) 6 d) 7 e) None of the above

34. $7^{12} - 4^{12}$ is exactly divisible by which of the following number?

a) 34 b) 33 c) 36 d) 35

35. If N, $(N + 2)$ and $(N + 4)$ are prime numbers, then the number of possible solutions for N are

a) 1 b) 2 c) 3 d) None of these

36. The smallest positive prime (say p) such that $2^p - 1$ is not a prime is _____

a) 5 b) 11 c) 17 d) 29

37. If b is the largest square divisor of c and a divides c, then which one of the following is correct? (where, a, b and c are integers)

a) b divides a b) a does not divide b

c) a divides b d) a and b are coprime

38. If n is a whole number greater than 1, then $n^2(n^2 - 1)$ is always divisible by

a) 12 b) 24 c) 48 d) 60 e) None of the above

39. What is the sum of all positive integers lying between 200 and 400 that are multiples of 7?

a) 8729 b) 8700 c) 8428 d) 8278

40. Consider the following statements

I. To obtain prime numbers less than 121, we are to reject all the multiples of 2, 3, 5 and 7.
II. Every composite number less than 121 is divisible by a prime number less than 11.

Which of the statements given above is/are correct?

a) Only I b) Only II c) Both I and II d) Neither I nor II

41. Consider the following statements

I. 7710312401 is divisible by 11.
II. 173 is a prime number.

Which of the statements given above is/are correct?

a) Only I b) Only II c) Both I and II d) Neither I nor II

42. If k is a positive integer, then every square integer is of the form

a) Only $4k$ b) $4k$ or $4k + 3$ c) $4k + 1$ or $4k + 3$ d) $4k$ or $4k + 1$

43. Every prime number of the form $3k + 1$ can be represented in the form $6m + 1$ (where k, m are integers) when

a) k is odd
b) k is even
c) k can be both odd and even
d) No such form is possible

Basic Skill Level Questions Answer Key													
1	c	6	a	11	a	16	c	21	b	26	c	31	d
2	a	7	d	12	d	17	b	22	d	27	a	32	c
3	c	8	b	13	d	18	a	23	d	28	b	33	b
4	a	9	a	14	c	19	d	24	b	29	a	34	c
5	d	10	c	15	a	20	b	25	a	30	c		

Higher Skill Level Questions																	
1	c	6	a	11	b	16	c	21	d	26	c	31	a	36	b	41	c
2	a	7	c	12	a	17	b	22	a	27	d	32	c	37	c	42	d
3	b	8	c	13	d	18	a	23	c	28	d	33	c	38	a	43	b
4	a	9	a	14	c	19	b	24	d	29	d	34	b	39	a		
5	a	10	d	15	c	20	c	25	a	30	d	35	a	40	c		

Chapter 2: Number Series

Directions: (Questions 1-30) Find the missing term in each of the following series.

1. 6, ?, 21, 33, 48

   a) 14 b) 20 c) 12 d) 18 e) None of the above

2. 4, 12, 36, ?, 324, 972

   a) 108 b) 125 c) 215 d) 312

3. 2, 8, 18, ? , 50, 72

   a) 33 b) 38 c) 36 d) 32

4. 10000, 2000, 400, 80, 16, 3.2, ?,

   a) 0.38 b) 0.45 c) 0.64 d) 0.54 e) None of the above

5. 2, 9, 28, 65, 126, ? ,

   a) 195 b) 199 c) 207 d) 217

6. 6, 13, 32, ?, 130, 221

   a) 75 b) 69 c) 100 d) 85 e) None of the above

7. . . . , 9, 25, 49, 81, 121

   a) 5 b) 1 c) 4 d) 3 e) None of the above

8. 3, 6, 12, 24, 48, 96, ?,

   a) 192 b) 182 c) 186 d) 198 e) None of the above

9. 12, 15, 21, 30, 42, 57, ?,

   a) 59 b) 63 c) 65 d) 75 e) None of the above

10. 3, 5, 9, 17, 33, 65, ? ,

    a) 129 b) 128 c) 126 d) 132 e) None of the above

11. In the following number series, one of the numbers is wrong. Find out the wrong number.

    14, 28, 112, 672, 5374, 53760

    a) 112 b) 672 c) 5374 d) 28 e) None of the above

12. 48, 23, ?, 4.25, 1.125

    a) 10.5 b) 10 c) 2.5 d) 11

13. 4, 25, 130, ?, 3280

a) 685 b) 645 c) 678 d) 655 e) None of the above

14. 41, 40, 36, ?, 11

a) 35 b) 27 c) 29 d) 30 e) None of the above

15. 135, 134, ?, 99, 35

a) 126 b) 115 c) 85 d) 111 e) None of the above

16. 1, 9, 25, 49, 81, ?

a) 90 b) 135 c) 121 d) 125 e) None of the above

17. 2, 15, 41, 80, 132, ?

a) 197 b) 150 c) 178 d) 180 e) None of the above

18. Find the missing number in the following series.

4, 18, . . . , 100, 180, 294

a) 32 b) 36 c) 48 d) 40

19. 840, ?, 420, 140, 35, 7

a) 408 b) 840 c) 480 d) 804 e) None of the above

20. 5, ?, 15, 75, 525, 4725, 51975

a) 5 b) 10 c) 8 d) 6 e) None of the above

21. 13, 35, 57, 79, 911, ?

a) 1113 b) 1123 c) 1114 d) 1124 e) None of the above

22. 125, 80, 45, ?, 5

a) 20 b) 30 c) 25 d) 35 e) None of the above

23. 4, 19, 49, ?, 229

a) 75 b) 109 c) 65 d) 169

24. Complete the series 7, 26, 63, 124, 215, 342, ? , [SSC CGL (Main) 2012]

a) 481 b) 511 c) 391 d) 421

25. 2, 6, 14, 30, . . . , 126

a) 62 b) 63 c) 73 d) 95

26. 165, 195, 255, 285, . . . , 435

a) 340 b) 341 c) 345 d) 401 e) None of the above

27. 1220, 244, . . . , 9.76, 1.952

a) 48.8 b) 49.2 c) 47 d) 8 e) None of the above

28. 2, 10, 42, 170, ?, 2730, 10922

a) 588 b) 568 c) 596 d) 682 e) None of the above

29. 23, 42.2, 80.6, 157.4, 311, ?

a) 618.2 b) 623.6 c) 624.2 d) 616.6 e) None of the above

30. 14, 33, 104, ?, 2110

a) 421 b) 433 c) 372 d) 840 e) None of the above

Directions (Q. Nos. 31-35) Find the wrong number in each of the following series.

31. 2, 11, 38, 197, 1172, 8227, 65806

a) 11 b) 38 c) 197 d) 1172 e) None of the above

32. 16, 19, 21, 30, 46, 71, 107

a) 19 b) 20 c) 30 d) 46 e) 71

33. 7, 9, 16, 25, 41, 68, 107, 173

a) 107 b) 16 c) 41 d) 68 e) 25

34. 4, 2, 3.5, 7.5, 26.25, 118.125,

a) 118.125 b) 26.25 c) 3.5 d) 2 e) 7.5

35. 16, 4, 2, 1.5, 1.75, 1.875

a) 1.875 b) 1.75 c) 1.5 d) 2 e) 4

36. The wrong number in the series

2, 9, 28, 65, 126, 216, 344

a) 9 b) 65 c) 216 d) 25

37. What should come in place of the question mark (?) in the following number series?

3, 3, 12, 108, ?, 42300

a) 2700 b) 1728 c) 972 d) 432 e) None of the above

38. Which of the following numbers does not fit into the series?

14, 19, 29, 40, 44, 51, 59, 73

a) 59 b) 51 c) 44 d) 29

Higher Skilled Level Questions

Directions(Q. Nos. 1-25) What should come in place of the question mark in each of the following number series?

1. 50, 60, 75, 97.5, ? 184.275, 267.19875

   a) 120.50 b) 130.50 c) 131.625 d) 124.25 e) None of the above

2. 12, 15, 36, ?, 480, 2415, 14508

   a) 115 b) 109 c) 117 d) 121 e) None of the above

3. 1, 2, 6, 21, 88, 445, ?

   a) 2230 b) 2676 c) 2580 d) 2670 e) None of the above

4. 20, 21, 25, 34, 50, ?, 111

   a) 70 b) 65 c) 60 d) 75 e) None of the above

5. 600, 125, 30, ?, 7.2, 6.44, 6.288

   a) 6 b) 10 c) 15 d) 11 e) None of the above

6. 12, 24, 72, 144, 432, ? ,

   a) 728 b) 852 c) 864 d) 1296 e) None of the above

7. 13, 16, 22, 33, 51, ?

   a) 89 b) 78 c) 102 d) 69 e) None of the above

8. 39, 52, 78, 117, 169, ?

   a) 246 b) 182 c) 234 d) 256 e) None of the above

9. 62, 87, 187, 412, 812, ?

   a) 1012 b) 1437 c) 1337 d) 1457 e) None of the above

10. 24, 536, 487, 703, 678, ?

    a) 768 b) 748 c) 764 d) 742 e) None of the above

11. 656, 432, 320, 264, 236, ?

    a) 222 b) 229 c) 232 d) 223 e) None of the above

12. 9, 62, ?, 1854, 7415, 22244

    a) 433 b) 309 c) 406 d) 371 e) None of the above

13. 4, 8, 24, 60, ?, 224

    a) 178 b) 96 c) 109 d) 124 e) None of the above

14. 36, 154, 232, 278, 300, ?

a) 306 b) 313 c) 308 d) 307 e) None of the above

15. 456.5, 407, 368.5, 341, 324.5, ? ,

a) 321 b) 319 c) 317 d) 323 e) None of the above

16. 7, 8, 18, ?, 232, 1165

a) 84 b) 42 c) 57 d) 36 e) None of the above

17. 5, 54, 90, 115, 131, 140, ?

a) 149 b) 146 c) 144 d) 152 e) None of the above

18. 7, 4, 5, 9, ?, 52.5, 160.5

a) 32 b) 16 c) 14 d) 20 e) None of the above

19. 6, 42, ?, 1260, 5040, 15120, 30240

a) 546 b) 424 c) 252 d) 328 e) None of the above

20. 4, 10, 40, 190, 940, ?, 23440

a) 4690 b) 2930 c) 5140 d) 3680 e) None of the above

21. 2, 9, 30, ?, 436, 2195, 13182,

a) 216 b) 105 c) 178 d) 324 e) None of the above

22. 3, 5, 7, ?, 13, 17, 19, 23

a) 9 b) 11 c) 8 d) 10 e) None of the above

23. Next number in the series 6, 16, 36, 76, 156, 316 is

a) 636 b) 638 c) 633 d) 632

24. 123, 277, 459, 669, 907, ?

a) 1179 b) 1173 c) 1167 d) 1169 e) None of the above

25. 150, 152, 149, 153, 148, 154, ?

a) 155 b) 152 c) 147 d) 149 e) None of the above

Directions (Q. Nos. 26-35) Find the wrong number in each of the following series.

26. 13, 25, 40, 57, 79, 103, 130

a) 25 b) 40 c) 57 d) 79 e) None of the above

27. 850, 600, 550, 500, 475, 462.5, 456.25

a) 600 b) 550 c) 500 d) 462.5 e) None of the above

28. 2, 10, 18, 54, 162, 486, 1458 [Bank PO 2010]

a) 18 b) 54 c) 162 d) 10 e) None of the above

29. 8, 12, 24, 46, 72, 108, 52

a) 12 b) 24 c) 46 d) 72 e) None of the above

30. 142, 119, 100, 83, 65, 59, 52

a) 65 b) 100 c) 59 d) 119 e) None of the above

31. 12, 6, 7.5, 12.75, 27.5, 71.25 [Bank PO 2011]

a) 6 b) 7.5 c) 27.5 d) 12 e) None of the above

32. 16, 24, 37, 54, 81, 121.5 [Hotel Mgmt. 2011]

a) 24 b) 54 c) 37 d) 121.5 e) None of the above

33. 12, 12, 18, 48, 180, 1080 [Bank Clerks 2009]

a) 180 b) 12 c) 18 d) 48 e) None of the above

34. 22, 23, 27, 36, 58, 77 [Bank Clerks 2008]

a) 77 b) 58 c) 36 d) 22 e) None of the above

35. 16, 14, 24, 66, 250, 1270

a) 250 b) 66 c) 16 d) 1270 e) None of the above

Directions (Q. Nos. 36-45) There are two series given below, of which one is complete and follows a certain pattern. Based on the pattern of complete series, you have to find the different answering terms of series.

36.

1275	1307	1371	1467	1595	1755
972	*A*	*B*	*C*	*D*	*E*

Which number will come at the place of D?

a) 1292 b) 550 c) 500 d) 462.5 e) None of the above

37.

4	23	113	449	1343	2681
7	*A*	*B*	*C*	*D*	*E*

Which number will come at the place of E?

a) 4793 b) 4782 c) 4841 d) 4932 e) None of the above

38.

3	4	10	33	136	685
7	*A*	*B*	*C*	*D*	*E*

Which number will come at the place of E?

a) 57 b) 1165 c) 18 d) 1398 e) None of the above

39.

6	9	18	45	135	472.5
20	*A*	*B*	*C*	*D*	*E*

Which number will come at the place of C?

a) 30 b) 1125 c) 375 d) 150 e) None of the above

40.

2	9	57	337	1681	6721
7	*A*	*B*	*C*	*D*	*E*

Which number will come from the following at place of E?

a) 673 b) 3361 c) 13441 d) 17 e) None of the above

41.

25	194	73	154	105
14	*A*	*B*	*C*	*D*

Which number will come from the following at place of D?

a) 183 b) 62 c) 69 d) 94 e) None of the above

42.

140	68	36	16	10	3
284	*A*	*B*	*C*	*D*	*E*

Which number will come from the following at place of B?

a) 72 b) 140 c) 34 d) 18 e) None of the above

43.

3	20	118	587	2344	7027
12	*A*	*B*	*C*	*D*	*E*

Which number will come from the following at place of C?

a) 2477 b) 83 c) 496 d) 9904

44.

4	16	48	120	272	584
124	*A*	*B*	*C*	*D*	*E*

Which number will come from the following at place of B?

a) 256 b) 528 c) 1080 d) 4424

45.

8	584	143	467	242	386
43	*A*	*B*	*C*	*D*	*E*

Which number will come from the following at place of C?

a) 321 b) 277 c) 519 d) 502 e) None of the above

Basic Skill Level Questions Answer Key															
1	c	6	b	11	c	16	c	21	a	26	c	31	d	36	c
2	a	7	b	12	a	17	a	22	a	27	a	32	a	37	b
3	d	8	a	13	d	18	c	23	b	28	d	33	d	38	b
4	c	9	d	14	b	19	b	24	b	29	a	34	c		
5	d	10	a	15	a	20	a	25	a	30	a	35	b		

Higher Skill Level Questions																	
1	c	6	c	11	a	16	c	21	b	26	c	31	a	36	a	41	d
2	c	7	b	12	d	17	c	22	b	27	a	32	c	37	c	42	a
3	b	8	c	13	d	18	d	23	a	28	d	33	d	38	b	43	a
4	d	9	b	14	a	19	c	24	b	29	c	34	b	39	d	44	b
5	d	10	d	15	b	20	a	25	c	30	a	35	a	40	e	45	d

Chapter 3: Simple and Decimal Fractions

1. Convert 0.34 into vulgar fraction.

   a) $\frac{18}{50}$ b) $\frac{16}{50}$ c) $\frac{17}{50}$ d) $\frac{19}{50}$ e) None of the above

2. Find the sum $\frac{3}{10} + \frac{5}{100} + \frac{8}{1000}$ in decimal form.

   a) 0.853 b) 0.358 c) 3.58 d) 8.35 e) None of the above

3. Change $\frac{75}{100}$ into decimal number.

   a) 0.075 b) 0.75 c) 0.0075 d) 7.5 e) None of the above

4. Express $0.\overline{5}$ as vulgar fraction.

   a) $\frac{5}{10}$ b) $\frac{5}{12}$ c) $\frac{5}{9}$ d) $\frac{5}{6}$ e) None of the above

5. Convert $0.2\overline{7}$ into vulgar fraction.

   a) $\frac{5}{18}$ b) $\frac{5}{19}$ c) $\frac{7}{19}$ d) $\frac{3}{17}$ e) None of the above

6. Find the value of $9.46\overline{7}$ in a vulgar fraction.

   a) $9\frac{421}{900}$ b) $9\frac{422}{900}$ c) $9\frac{435}{900}$ d) $9\frac{437}{900}$ e) None of the above

7. Express $0.82\overline{68}$ as a vulgar fraction.

   a) $\frac{3093}{4950}$ b) $\frac{3043}{4850}$ c) $\frac{4093}{4950}$ d) $\frac{3039}{4950}$ e) None of the above

8. Arrange $\frac{7}{12}, \frac{2}{3}$ and $\frac{3}{8}$ in ascending order.

   a) $\frac{3}{8} < \frac{7}{12} < \frac{2}{3}$ b) $\frac{2}{3} < \frac{7}{12} < \frac{3}{8}$ c) $\frac{7}{12} < \frac{2}{3} < \frac{3}{8}$

   d) $\frac{3}{8} < \frac{2}{3} < \frac{7}{12}$ e) None of the above

9. Arrange $\frac{7}{9}, \frac{5}{8}$ and $\frac{3}{7}$ in descending order.

a) $\frac{7}{9} > \frac{5}{8} > \frac{3}{7}$ b) $\frac{3}{7} > \frac{5}{8} > \frac{7}{9}$ c) $\frac{5}{8} > \frac{7}{9} > \frac{3}{7}$

d) $\frac{3}{7} > \frac{7}{9} > \frac{5}{8}$ e) None of the above

10. Out of the fractions $\frac{5}{7}$, $\frac{7}{13}$, $\frac{4}{7}$, $\frac{4}{15}$ and $\frac{9}{14}$, which is the third highest?

a) $\frac{5}{7}$ b) $\frac{7}{13}$ c) $\frac{4}{7}$ d) $\frac{4}{15}$ e) $\frac{9}{14}$

11. Out of fractions $\frac{4}{7}$, $\frac{5}{13}$, $\frac{6}{11}$, $\frac{3}{5}$ and $\frac{2}{3}$, which is second smallest fraction.

a) $\frac{4}{7}$ b) $\frac{5}{13}$ c) $\frac{6}{11}$ d) $\frac{3}{5}$

12. 1088.88 + 1800.08 + 1880.80 = ?

a) 8790.86 b) 8890.86 c) 5588.80 d) 4769.76 e) None of the above

13. 6435.9 + 7546.4 + 1203.5 = ?

a) 15188.5 b) 15185.8 c) 15155.5 d) 15815.8 e) None of the above

14. 726.34 + 888.12 - ? = 1001.88

a) 612.58 b) 602.64 c) 654.54 d) 618.78 e) None of the above

15. $\frac{16}{23} \times \frac{47}{288} \times \frac{92}{141} = ?$

a) $\frac{4}{27}$ b) $\frac{2}{27}$ c) $\frac{2}{29}$ d) $\frac{3}{28}$ e) None of the above

16. $1\frac{3}{5} + 1\frac{8}{9} + 2\frac{4}{5} = ?$

a) $6\frac{19}{45}$ b) $6\frac{16}{45}$ c) $6\frac{17}{45}$ d) $6\frac{13}{45}$ e) None of the above

17. $\frac{19999}{21111} = ?$

a) 0.947 b) 0.749 c) 0.497 d) 0.794 e) 0.974

18. $\frac{1212}{0.5} = 6.06 \times ?$

a) 4.04 b) 400 c) 0.4 d) 0.44

19. $33 + 371 \div 7 = ?$

a) 89 b) 85 c) 86 d) 84 e) None of the above

20. $(39.3 \times 53.4) + (26.7 \times 5.9) = ?$

a) 2520.15 b) 2256.15 c) 2562.15 d) 2652.15 e) None of the above

21. $\frac{3}{4}$ of $\frac{5}{6}$ of $\frac{7}{10}$ of 1664 =?

a) 648 b) 762 c) 612 d) 728 e) None of the above

22. If the fractions $\frac{19}{21}$, $\frac{21}{25}$, $\frac{25}{29}$, $\frac{29}{31}$ and $\frac{31}{37}$ are arranged in ascending order of their values, then which one will be the 2nd?

a) $\frac{19}{21}$ b) $\frac{21}{25}$ c) $\frac{25}{29}$ d) $\frac{29}{31}$ e) None of the above

23. $783 \div 9 \div 0.75 = ?$

a) 130 b) 124 c) 118 d) 116 e) None of the above

24. $8\frac{1}{3} + 10\frac{5}{6} = ?$

a) $\frac{5}{6}$ b) $\frac{2}{3}$ c) $\frac{10}{13}$ d) $\frac{11}{13}$ e) None of the above

25. $0.4777\cdots =$ is equal to

a) $\frac{477}{10000}$ b) $\frac{477}{100}$ c) $\frac{437}{100}$ d) $\frac{43}{90}$ e) None of the above

26. 4+4.44+44.4+4.04+444=?

a) 472.88 b) 495.22 c) 500.88 d) 577.2 e) None of the above

27. $(0.\overline{142857}) \div (0.\overline{285714})$

a) $\frac{1}{2}$ b) $\frac{1}{3}$ c) 2 d) 10 e) None of the above

28. When 0.252525... is converted into a fraction, then find result?

a) $\frac{25}{99}$ b) $\frac{25}{90}$ c) $\frac{25}{999}$ d) $\frac{25}{9999}$

29. If $\frac{1}{1\frac{1}{4}} + \frac{1}{6\frac{2}{3}} - \frac{1}{x} + \frac{1}{10} = \frac{1}{12}$, then find the value of x

a) $\frac{15}{3}$ b) $\frac{20}{15}$ c) $\frac{15}{2}$ d) $\frac{12}{13}$ e) None of the above

30. The value of $0.\overline{3} + 0.\overline{6} + 0.\overline{7} + 0.\overline{8}$ in fraction will be

a) $2\frac{3}{10}$ b) $2\frac{2}{3}$ c) $20.\overline{35}$ d) $5\frac{3}{10}$ e) None of the above

31. Find the value of $999\frac{1}{7} + 999\frac{2}{7} + 999\frac{3}{7} + 999\frac{4}{7} + 999\frac{5}{7} + 999\frac{6}{7}$.

a) 5997 b) 5979 c) 5994 d) 2997

32. Find the value of $\frac{1}{3} + \frac{1}{15} + \frac{1}{35} + \frac{1}{63} + \frac{1}{99}$

a) $\frac{10}{11}$ b) $\frac{5}{11}$ c) $\frac{9}{11}$ d) $\frac{7}{11}$

33. $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{7} + \frac{1}{14} + \frac{1}{28} = ?$

a) 2 b) 5 c) 4 d) 6 e) None of the above

34. Find the value of $27 \times 1.\overline{2} \times 5.526\overline{2} \times 0.\overline{6}$.

a) $121.\overline{57}$ b) $121.\overline{75}$ c) $121.75\overline{5}$ d) None of the above

35. Out of the fractions $\frac{5}{7}, \frac{4}{9}, \frac{6}{11}, \frac{2}{5}$ and $\frac{3}{4}$, what is the difference between the largest and smallest fraction?

a) $\frac{6}{13}$ b) $\frac{11}{18}$ c) $\frac{7}{18}$ d) $\frac{11}{20}$ e) None of the above

36. If the numerator of a fraction is increased by 200% and denominator of the fraction is increased by 150%, the resultant fraction is $\frac{9}{35}$. What is the original fraction?

a) $\frac{3}{10}$ b) $\frac{2}{15}$ c) $\frac{3}{16}$ d) $\frac{2}{7}$ e) None of the above

37. $\frac{5}{12}$ part of what amount will be equal to $3\frac{3}{4}$ part of ₹100.

a) ₹900 b) ₹940 c) ₹875 d) ₹975

38. A, B, C and D purchase a gift worth ₹60. A pays 1/2 of what others are paying, B pays 1/3rd of what others are paying and C pays 1/4th of what others are paying. What is the amount paid by D?

a) 14 b) 15 c) 16 d) 13

39. 1/4th of number of boys and 3/8th of number of girls participated in annual sports of the school. What fractional part of total number of students participated?

a) 32% b) 20% c) 36% d) Data inadequate e) None of the above

40. The numerator of a fraction is 4 less than its denominator. If the numerator is decreased by 2 and the denominator is increased by 1, then the denominator becomes eight times the numerator, then find the fraction.

a) $\frac{3}{7}$ b) $\frac{4}{8}$ c) $\frac{2}{7}$ d) $\frac{3}{8}$

41. The greatest among the numbers $\sqrt[4]{2}$, $\sqrt[5]{3}$, $\sqrt[10]{6}$ and $\sqrt[20]{15}$ is

    a) $\sqrt[20]{15}$ b) $\sqrt[4]{2}$ c) $\sqrt[5]{3}$ d) $\sqrt[10]{6}$

42. If the fraction a/b is positive, then which of the following must be true?

    a) $a > 0$ b) $b > 0$ c) $ab > 0$ d) $a + b > 0$

Exercise: Higher Skill Level Questions

1. If $1.5x = 0.04y$, then find the value of $\frac{y-x}{y+x}$.

    a) $\frac{730}{77}$ b) $\frac{73}{77}$ c) $\frac{73}{770}$ d) $\frac{703}{77}$

2. If $0.764y = 1.236x$, then what is the value of $\frac{y-x}{y+x}$?

    a) 0.764 b) 0.236 c) 2 d) 0.472

3. Find the HCF of $\frac{5}{6}, \frac{15}{8}$ and $\frac{25}{32}$

    a) 0.052 b) 0.698 c) 0.75 d) Cannot be determined

4. Find the LCM of the fractions in above question.

    a) 13.5 b) 37.5 c) 10.5 d) Cannot be determined e) None of the above

5. The pair of rational numbers that lies between $\frac{1}{4}$ and $\frac{3}{4}$ is

    a) $\frac{262}{1000}, \frac{752}{1000}$ b) $\frac{24}{100}, \frac{74}{100}$ c) $\frac{9}{40}, \frac{31}{40}$ d) $\frac{252}{1000}, \frac{748}{100}$

6. 1/8 part of a pencil is black and 1/2 part of the remaining is white. If the remaining part is blue and length of this blue part is $3\frac{1}{2}$ cm, then find the length of the pencil.

    a) 6 *cm* b) 7 *cm* c) 8 *cm* d) 9 *cm* e) None of the above

7. How many pieces of 13.2 *cm* can be cut from a 330 *cm* long rod?

    a) 25 b) 28 c) 21 d) 35 e) None of the above

8. In the year 2011, Shantanu gets ₹3832.5 as his pocket allowance. Find his pocket allowance per day.

    a) ₹9.5 b) ₹10.5 c) ₹12.5 d) ₹11.5 e) None of the above

9. Sum of three fractions is $2\frac{11}{24}$. If the greatest fraction is divided by the smallest fraction, the result is 7/6, which is greater than the middle fraction by 1/3. Find all the three fractions.

    a) $\frac{3}{5}, \frac{4}{7}, \frac{2}{3}$ b) $\frac{7}{8}, \frac{5}{6}, \frac{3}{4}$ c) $\frac{7}{9}, \frac{2}{3}, \frac{3}{5}$ d) $\frac{7}{8}, \frac{7}{9}, \frac{7}{10}$ e) None of the above

10. Find the value of $2 \times \left(\dfrac{3.6 \times 0.48 \times 2.50}{0.12 \times 0.09 \times 0.5} \right)$

    a) 800 b) 500 c) 900 d) 1600 e) None of the above

11. If a fraction is multiplied by itself and then divided by the reciprocal of the same fraction, the result is $18\frac{26}{27}$. Find the fraction.

    a) $\dfrac{8}{27}$ b) $1\dfrac{1}{3}$ c) $2\dfrac{2}{3}$ d) $3\dfrac{2}{3}$ e) None of the above

12. What is the value of $0.007 + 0.00\overline{7} + 17.\overline{83} + 310.020\overline{2}$?

    a) 327.86638 b) 328.644 c) 327.86683 d) $327.866\overline{8}$

13. Representation of $0.2\overline{341}$ in the form p/q where p and q are integers, $q \neq 0$, is

    a) $\dfrac{781}{3330}$ b) $\dfrac{1171}{4995}$ c) $\dfrac{2341}{9990}$ d) $\dfrac{2339}{9990}$

14. If $1\dfrac{2}{3} \div \dfrac{2}{7} \times \dfrac{x}{7} = 1\dfrac{1}{4} \times \dfrac{2}{3} \div \dfrac{1}{6}$, then find the value of x.

    a) 0.006 b) $\dfrac{1}{6}$ c) 0.6 d) 6 e) None of the above

15. $\dfrac{0.04}{0.03}$ of $\dfrac{\left(3\dfrac{1}{3} - 2\dfrac{1}{2}\right) \div \dfrac{1}{2} \text{ of } 1\dfrac{1}{4}}{\dfrac{1}{3} + \dfrac{1}{5} \text{ of } \dfrac{1}{9}}$

    a) 1 b) 5 c) $\dfrac{1}{5}$ d) $\dfrac{1}{2}$

16. A man reads 3/8 of a book on a day and 4/5 of the remainder on the second day. If the number of pages still unread are 40, then how many pages did the book contain?

    a) 300 b) 500 c) 320 d) 350 e) None of the above

17. 4/7 of a pole is in the mud. When 1/3 of it is pulled out, 250 *cm* of the pole is still in the mud. Find the full length of the pole.

    a) 1000 b) 1100 c) 950 d) 1050 e) None of the above

Basic Skill Level Questions Answer Key																
1	c	6	a	11	c	16	d	21	d	26	c	31	a	36	e	41	c
2	b	7	c	12	d	17	a	22	b	27	a	32	b	37	a	42	c
3	b	8	a	13	b	18	b	23	d	28	a	33	a	38	d		
4	c	9	a	14	a	19	c	24	c	29	c	34	d	39	d		
5	a	10	c	15	b	20	b	25	d	30	b	35	e	40	a		

Higher Skill Level Questions							
1	b	6	c	11	c	16	c
2	b	7	a	12	b	17	d
3	a	8	b	13	d		
4	b	9	b	14	d		
5	d	10	d	15	b		